Abhishake Kundu

Research Associate
School of Industrial Engineering, Pontificia Universidad Católica de Valparaíso
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Research Interests

Large-scale energy systems modeling; Transport sector decarbonization; Combinatorial optimization for multi-modal vehicle systems; Emerging technologies in logistics; Development of efficient and scalable optimization problems and heuristics; Game theoretic models and bi-level programming; Energy markets; Machine learning and big data analysis (energy, healthcare); Statistical analysis (parametric and non-parametric).

Education

Ph.D., Industrial, Manufacturing and Systems Engineering; Texas Tech University, Lubbock, TX; May 2022
Kundu, A. (2022). *Developing efficient order and split heuristics for coordinated covering tour problems with drones.* (Doctoral dissertation)
Research Exchange Student, School of Industrial Engineering; Pontificia Universidad Católica de Valparaíso, Chile; Jan 2019

M.S., Industrial, Manufacturing and Systems Engineering; Texas Tech University, Lubbock, TX; Dec 2016
GPA: 4.0 / 4.0

B.Tech., Applied Electronics and Instrumentation Engineering; Heritage Institute of Technology, Kolkata, India; May 2011
GPA: 8.21 / 10

Academic Appointments

Pontificia Universidad Católica de Valparaíso, Chile
Research Associate, Apr 2026 – Present
Research: Long-term energy modeling, Game theoretic modeling, Transport sector decarbonization, Emerging technologies in logistics
Lectures: EII808 Programación Matemática, EII802 Modelos Estocásticos (Ph.D. Level); EII4220-1/2 Introducción a Data Science (Undergraduate Level)

Postdoctoral Researcher and Lecturer, Jul 2023 – Mar 2026
Research: Long-term energy modeling, Game theoretic modeling
Lectures: EII808 Programación Matemática, EII802 Modelos Estocásticos (Ph.D. Level); EII4220-1/2 Introducción a Data Science (Undergraduate Level)

Texas Tech University, Lubbock, TX
Graduate Part-time Instructor/ Teaching/ Research Assistant, Sep 2016 – Oct 2021
Optimization: IE 3311 Deterministic Operations Research (Undergraduate Level); IE 5318 Operations Research Modeling with Spreadsheets (Graduate Level)
Engineering Statistics: IE 2341 Engineering Statistics I, IE 3342 Engineering Statistics II, IE 4325 Productivity and Performance Improvement in Organizations: Lean Methods (Under-

graduate Level); IE 5342 Design of Experiments, IE 5344 Statistical Data Analysis (Graduate Level)

Simulation/Programming: IE 2311 Computing for Industrial Engineers, IE 4316 Simulation Systems Modeling (Undergraduate Level); IE 5316 Simulation Models for Operations Analysis (Graduate Level)

Publications

- Kundu, A., Gatica-Escobar, R., & Matis, T. I. (2022). An efficient routing heuristic for a drone-assisted delivery problem. *IMA Journal of Management Mathematics*, **33**(4), 583–601. https://doi.org/10.1093/imaman/dpab039

- Kundu, A., Feijoo, F., Martinez, D., Hermosilla, M., & Matis, T.I. (2022). Prospective adverse event risk evaluation in clinical trials. *Health Care Management Science*, **25**(1), 89–99. https://doi.org/10.1007/s10729-021-09584-y

- Kundu, A., Feijoo, F., Mesa, F., Sankaranarayanan, S., Aristizábal, A. J., & Castaneda, M. (2023). Power on the go: A solution to address electric vehicle charging challenges. *Mathematics*, **12**(1), 91. https://doi.org/10.3390/math12010091

- Feijoo, F., Kundu, A., Flores, F., & Matamala, Y. (2024). Photovoltaic sizing assessment for microgrid communities under load shifting constraints and endogenous electricity prices: A Stackelberg approach. *Energy*, **307**, 132758. https://doi.org/10.1016/j.energy.2024.132758

- Kundu, A., Flores, F., Fernández Vázquez, C. A. A., Godoy, H., Quoilin, S., Castaneda, M., & Feijoo, F. (2025). Evaluating the impact of flexibilities from heating and electromobility in Chile's carbon neutrality pathway. *Buildings*, **15**(1), 68. https://doi.org/10.3390/buildings15010068

- Feijoo, F., Flores, F., Kundu, A., Pfeifer, A., Herc, L., Prieto, A. L., & Duic, N. (2025). Tradeoffs between economy wide future net zero and net negative economy systems: The case of Chile. *Renewable and Sustainable Energy Reviews*, **207**, 114945. https://doi.org/10.1016/j.rser.2024.114945

- Lillo Otarola, L., de la Fuente-Mella, H., Peña Domarchi, A., Kundu, A., & Ceroni-Díaz, J. (2025). Optimal scheduling of the wine-bottling process: A multi-dependency model with hydraulic considerations. *Applied Sciences*, **15**(9), 4697. https://doi.org/10.3390/app15094697

- Schaumann, S., Kundu, A., Pina-Pardo, J.C., Winkenbach, M., Gatica, R., Wagner, S.M., & Matis, T.I. (2025). The Flying Sidekick Traveling Salesman Problem with Multiple Drops: A Simple and Effective Heuristic Approach. *Computers & Industrial Engineering*, **210**, 111523. https://doi.org/10.1016/j.cie.2025.111523

- Oyarzún, F., Feijoo, F., Kundu, A., & Castaneda, M. (2026). Assessing policy strategies for residential solar photovoltaic adoption: A system dynamics approach with heuristic optimization. *Renewable Energy*, **274**, 126127. https://doi.org/10.1016/j.renene.2026.126127

- Pastore, L. M., Groppi, D., Feijoo, F., Flores, F., Kundu, A., Herc, L., Astiaso Garcia, D., de

Santoli, L., & Duić, N. (2026). Assessing the future cost of hydrogen in a fully decarbonised Italian energy system: a comparison of myopic and perfect foresight optimisation approaches. *International Journal of Hydrogen Energy*, **268**, 157058. https://doi.org/10.1016/j.ijhydene.2026.157058

- Herc, L., Kundu, A., Ali, S., Feijoo, F., Pfeifer, A., & Duić, N. (2026). Implementation of the myopic approach in the long-term energy planning and capacity expansion model; examination in a multizonal case. *Energy Conversion and Management*, **370**, 122178. https://doi.org/10.1016/j.enconman.2026.122178

- Kundu, A., & Feijoo, F. (2025). A study on coalition formation and bargaining power for the North American natural gas market. *Utilities Policy* (under second review). http://dx.doi.org/10.2139/ssrn.5442696

- Feijoo, F., Kundu, A., Flores, F., & Paredes, P. (2025). Energy community strategies based on heat and electricity demand response and technology efficiency: A Stackelberg approach. *Renewable Energy* (under second review).

- Kundu, A., Flores, F., Soto Morales, I., Villanueva García, I., Loaiza González, I., Khan, M.M.K., & Feijoo, F. (2026). A comparative assessment of hydrogen certification in APEC economies: Maturity, design, and stringency. *Renewable and Sustainable Energy Reviews* (submitted).

Working Papers

- Kundu, A., Starčević, M., Feijoo, F., & Duić, N. Study on the power-to-vehicle flexibilities in the H2RES model under endogenous pricing and battery degradation. (Target submission: Dec 2026)

- Kundu, A., Prieto, D., & Aros, D. Quantifying the cost of priorities in home healthcare routing: A lexicographic MILP study. (Target submission: Mar 2027)

Technical Reports

- Feijoo, F., Kundu, A., Flores, F., Baeriswyl Concha, A., Soto Morales, I., & Villanueva García, I. (2026). *Certification of Hydrogen and its Derivatives in APEC Economies: Its Role in Driving the Market*. APEC Project EWG 105 2024A. APEC Publication on Hydrogen Certification

- Feijoo, F., Kundu, A., Flores, F., Paredes, P., Soto Morales, I., & Villanueva García, I. (2026). *Building Resilient Energy Policies in Asia-Pacific*. APEC Project EWG 104 2024A. APEC Publication on Resilient Energy Policies

- Feijoo, F., Kundu, A., Flores, F., & Paredes, P. (2026). Section 3.3: The role of energy decarbonization in the water-energy-food nexus. In *Thematic Report. Water-Energy-Food Nexus in Chile: Science and Policies for Sustainable Management*. Comité Científico Asesor de Cambio Climático, Chile. https://doi.org/10.5281/zenodo.20071201

- Feijoo, F., Flores, F., Gonzalez, A., Paredes, P., Kundu, A., & Puin, D. *Local Demand as an Enabler to a Low-Carbon Hydrogen Economy*. APEC Project EWG 105 2025A (forthcoming, 2027).

Conference Presentations

- Kundu, A. & Matis, T.I. (2017). A Delivery Time Reduction Heuristic using Drones under Windy Conditions. *Institute of Industrial and Systems Engineering Annual Conference*, Pittsburgh, Pennsylvania.

- Kundu, A. & Matis, T.I. (2017). Establishing Energy Efficient Paths for Delivery Drones in Direction Dependent Uniform Wind-fields. *Institute for Operations Research and Management Science Annual Conference*, Houston, Texas.

- Kundu, A., Gatica-Escobar, R., & Matis, T.I. (2018). Integrated Covering & Routing Problems: Applications in Drone Assisted Logistics Operations. *Institute for Operations Research and Management Science Annual Conference*, Phoenix, Arizona.

- Kundu, A., Feijoo, F., Martinez, D., Hermosilla, M., & Matis, T.I. (2018). Assessment of Risk Based Metrics for Adverse Events in Oncology Clinical Trials. *III International Workshop on Data Science*, Valparaíso, Chile.

- Kundu, A., Gatica-Escobar, R., & Matis, T.I. (2020). An Efficient Order and Split Heuristic for Drone Assisted Delivery Problems. *Institute for Operations Research and Management Science Annual Conference*, Online.

- Kundu, A., Gatica-Escobar, R., & Matis, T.I. (2021). Developing an Efficient Routing Heuristic for a Drone-Assisted Delivery Problem. *Institute for Operations Research and Management Science Annual Conference*, Anaheim, California.

- Kundu, A. & Feijoo, F. (2024). Power on the Go: A Multi-objective Approach to Ease Electric Vehicle Charging Challenges. *Conference on Sustainable Development of Energy, Water and Environment Systems*, Rome, Italy.

- Kundu, A., Starcevic, M., Feijoo, F., & Duic, N. (2025). Study on the Power-to-Vehicle Flexibilities in the H2RES Model under Endogenous Pricing and Battery Degradation. *Conference on Sustainable Development of Energy, Water and Environment Systems*, Dubrovnik, Croatia.

- Kundu, A., Flores, F., Loaiza González, I., Khan, M.M.K., & Feijoo, F. (2026). Current Status and Comparative Insights of Hydrogen Certification in APEC Economies. *Conference on Sustainable Development of Energy, Water and Environment Systems*, Cartagena, Colombia.

Funded Research

NSF Award #2124331, Senior Personnel (2021)

I-Corps: Assessment Platform for Operationalizing High Reliability Organizational Hallmarks in Healthcare. Funding: $50,000

Summary: The HRO safety management model uses text and data mining techniques and was empirically developed from organizations in high-consequence, low-risk industries that consistently avoid failure. It uses near-miss and related data to help healthcare organizations assess and improve safety performance and generate risk scores.

Awards & Honors

- **Industrial Engineering Graduate Scholarship**, Department of Industrial Engineering, Texas Tech University (2014–2016).
- **Best Presenter Award**, III Workshop on Data Science, Valparaíso, Chile (2018).

Academic Service & Outreach

- **Scientific Advisory Board**, Conference on Sustainable Development of Energy, Water and Environment Systems (SDEWES): 19th SDEWES, Rome, Italy (2024); 20th SDEWES, Dubrovnik, Croatia (2025); 1st African SDEWES, Oujda-Saidia, Morocco (2025); 21st SDEWES, Gran Canaria, Spain (2026); 5th Latin American SDEWES, Cartagena, Colombia (2026); 2nd African SDEWES, Dar es Salaam, Tanzania (2027).
- **Session Chair**, Conference on Sustainable Development of Energy, Water and Environment Systems (SDEWES): 19th SDEWES (Rome, 2024) — *Smart Energy Systems*; 20th SDEWES (Dubrovnik, 2025) — *Energy System Analysis 7*.
- **Local Organizing Committee**, 4th Latin American SDEWES Conference, Viña del Mar, Chile (2024).
- **Peer Reviewer**, *Energy* (Elsevier, journal publications) and *Computers & Operations Research* (Elsevier, journal publications); Agencia Nacional de Investigación y Desarrollo, Chile (funding proposal).
- **Instructor and Presenter**, Explore Engineering Summer Camp (for high-school female students in STEM), Texas Tech University (2018). Topic: *Impact of Lean—From Manufacturing Processes to Everyday Life*.

Technical Skills

Python, Java, R, C++, CPLEX, Gurobi, ARENA, Simio, Mathematica, Matlab, LaTeX, H2RES Core Developer (an open-source, sectorally-integrated long-term energy planning model)

Industry Experience

Total Instrumentation Services, Cochin, India
Site Project Engineer (Instrumentation), Feb 2013 – Mar 2014

Phillips Carbon Black Ltd., Cochin, India
Assistant Manager (Instrumentation), Jul 2011 – Dec 2012

Languages

English; Bengali; Hindi, Spanish (moderate)

References

- **Dr. Timothy I. Matis**
  Associate Professor, Department of Industrial, Manufacturing and Systems Engineering, Texas Tech University, USA
  E-mail: timothy.matis@ttu.edu | Phone: +1 806 742 3543

- **Dr. Ricardo Gatica Escobar**
  Professor, School of Industrial Engineering, Pontificia Universidad Católica de Valparaíso, Chile
  E-mail: ricardo.gatica@pucv.cl | Phone: +56 32 227 3707

- **Prof. Masud Khan**
  Professor, School of Engineering, Computer and Mathematical Sciences, Auckland University of Technology, New Zealand
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- **Dr. Felipe Feijoo Palacios**
  Associate Professor, School of Industrial Engineering, Pontificia Universidad Católica de Valparaíso, Chile
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- **Dr. Jennifer Cross**
  Associate Professor, Department of Industrial, Manufacturing and Systems Engineering, Texas Tech University, USA
  E-mail: jennifer.cross@ttu.edu | Phone: +1 806 742 3543

- **Prof. Neven Duić**
  Chair, Faculty of Mechanical Engineering and Naval Architecture, University of Zagreb, Croatia
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